



A spatial perspective on the stability-complexity relationship

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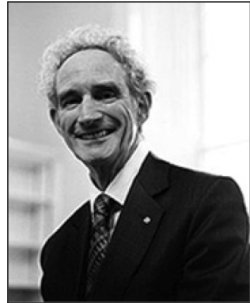
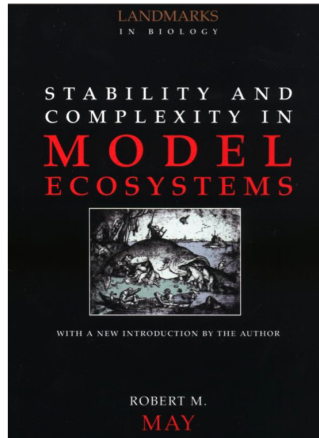


An old and intuitive interpretation

For decades, a dominant paradigm was that complex communities are more stable (Odum 1953; MacArthur 1955; Elton 1958; Hutchinson 1959).

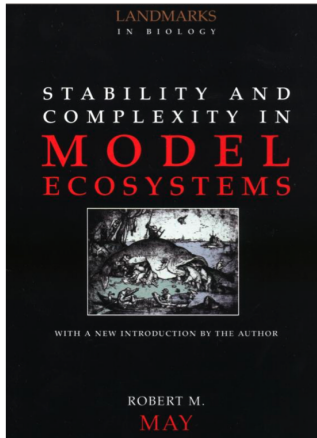
MacArthur (1955) stated very generally that "a larger number of paths through each species is necessary to reduce the effects of overpopulation of one species". He later concluded that "stability increases as the number of links increases".

But....





Stability of random ecosystems



$$J = \begin{bmatrix} - & + & 0 & 0 & - \\ 0 & - & 0 & 0 & 0 \\ 0 & - & - & 0 & 0 \\ - & 0 & 0 & - & - \\ 0 & 0 & + & 0 & - \end{bmatrix}$$



Stability of random ecosystems

Criteria for local stability of complex random ecosystems:

$$\alpha\sqrt{SC} < 1$$

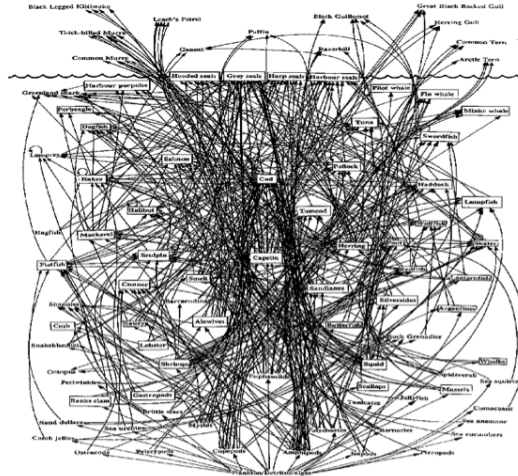
Where:

- ▶ α : standard deviation of non null elements of the Jacobian
- ▶ S : species richness
- ▶ C : Connectance

$$J = \begin{bmatrix} - & + & 0 & 0 & - \\ 0 & - & 0 & 0 & 0 \\ 0 & - & - & 0 & 0 \\ - & 0 & 0 & - & - \\ 0 & 0 & + & 0 & - \end{bmatrix}$$



The complexity paradox





Food webs are not random

- Pimm, S.L. (1982). Food Webs.
University of Chicago Press. Chicago.
- ▶ Food webs are not random
 - ▶ There are multiple meanings to stability





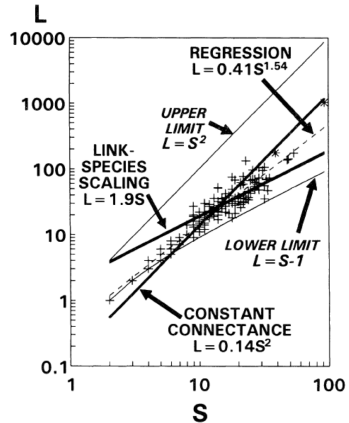
Testing scaling relationships

Given the relationship

$$\alpha\sqrt{SC} = \alpha\sqrt{\frac{L}{S}} < 1$$

We should find a relationship between L and S , such that:

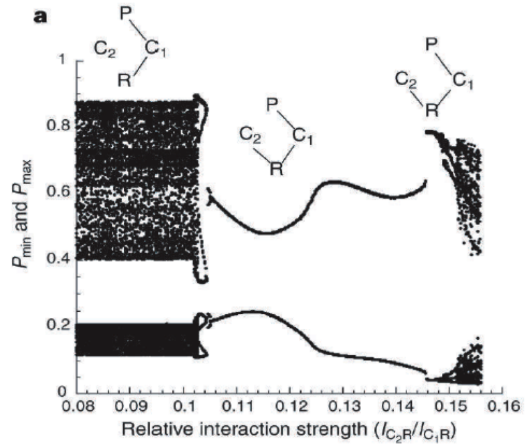
$$\log(L) = \frac{1}{2}\log(\alpha) + \log(S)$$





And interactions are not normally distributed

What about the α parameter?

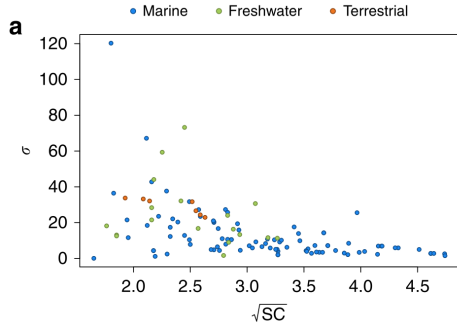


McCann et al. 1998. Nature.



And interactions are not normally distributed

What about the α parameter?



Jacquet et al. 2016. Nat. Communications

But what about spatial dynamics ?



Rosenzweig-MacArthur consumer-resource model

Resource dynamics follow a logistic growth, minus consumption :

$$\frac{dR}{dt} = rR * \left(1 - \frac{R}{K}\right) - \frac{aRC}{1 + bR}$$

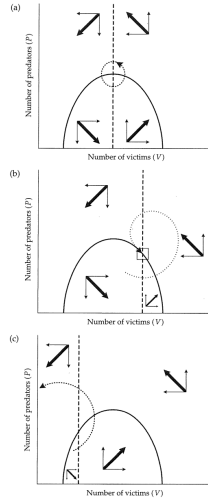
And consumer dynamics simply transform consumption into biomass, minus mortality :

$$\frac{dC}{dt} = \frac{eaRC}{1 + bR} - mC$$

You can solve the equation at equilibrium by setting differential equations to 0 and isolating the two variables.



Rosenzweig-MacArthur predator-prey model Paradox of enrichment





Rosenzweig-MacArthur predator-prey model Adding resource immigration

Resource immigration is stabilizing because it induces indirect density-dependence :

$$\frac{dR}{dt} = rR * \left(1 - \frac{R}{K}\right) - \frac{aRC}{1 + bR} + I$$

Or :

$$\frac{dR}{dt} \frac{1}{R} = r\left(1 - \frac{R}{K}\right) - \frac{aC}{1 + bR} + \frac{I}{R}$$



Rosenzweig-MacArthur predator-prey model Adding consumer immigration

In the situation where :

$$\frac{dC}{dt} = \frac{eaRC}{1 + bR} - mC + I$$

And we find that increasing I

- ▶ Drives the resource towards extinction
- ▶ Generates indirect density-dependence, which is also stabilizing



Rosenzweig-MacArthur predator-prey model Discrete landscape

Consider a two patch system :

$$\frac{dR_x}{dt} = rR_x * \left(1 - \frac{R_x}{K}\right) - \frac{aR_x C_x}{1 + bR_x} - d_R R_x + d_R (R_{x-1} + R_{x+1})/2$$

And

$$\frac{dC_x}{dt} = \frac{eaR_x C_x}{1 + bR_x} - mC_x - d_C C_x + d_c (C_{x-1} + C_{x+1})/2$$

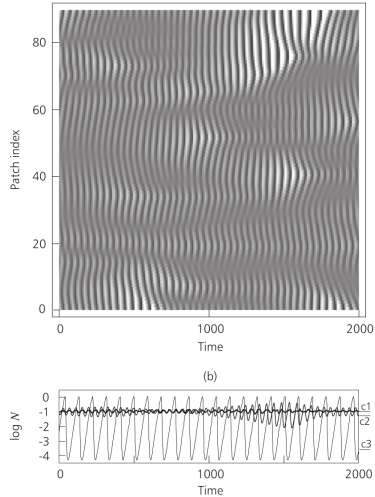


Rosenzweig-MacArthur predator-prey model Discrete landscape

- ▶ Implement the model over a large one dimensional lattice, without diffusion
- ▶ Extract and plot time series
- ▶ Tune parameters to get close to instability
- ▶ Add progressively the resource and consumer diffusion parameters
- ▶ **Advanced** : add spatial variability in K and compare results



From local to regional dynamics

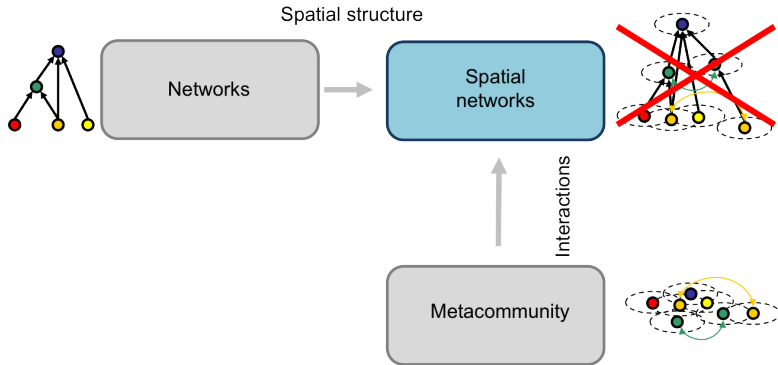


Jansen et al. 2000

*Stability and complexity in random
metaecosystems*



From local to regional dynamics

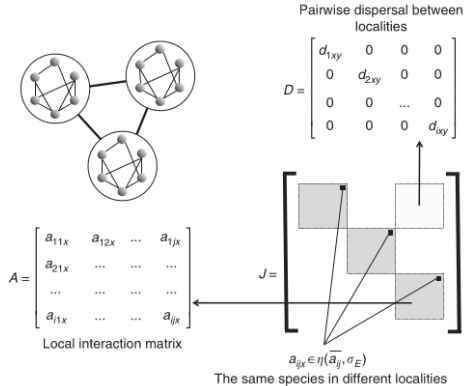




Jacobian matrices in space

What's happening with spatialization?

- ▶ The number of equations increases
- ▶ Connectance decreases
- ▶ We don't know what's happening with α



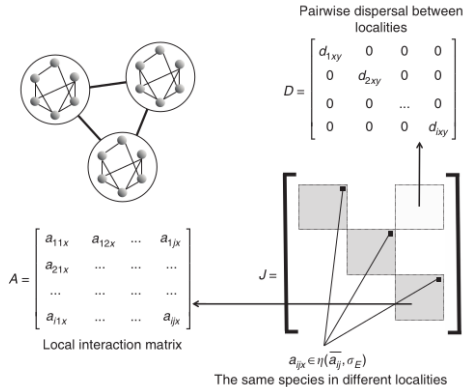
Gravel et al. 2016. Nat. Communications



Jacobian matrices in space

Two problems:

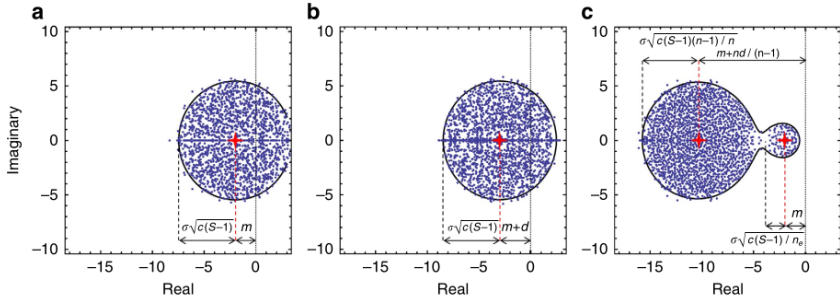
1. Stability of the equilibrium
2. Feasibility of the equilibrium



Gravel et al. 2016. Nat. Communications



Analytical results



Gravel et al. 2016. Nat. Communications

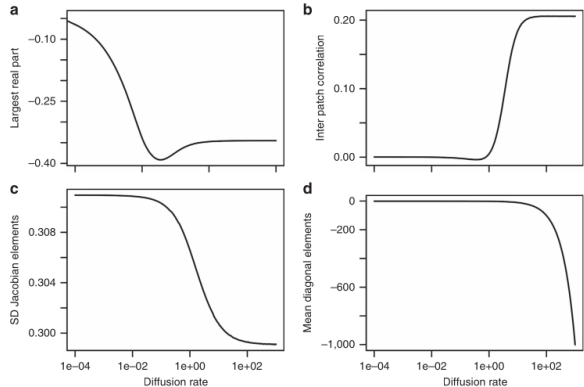
Spatial structure and diffusion

- Move the center of the disk toward the left
- Reduce the radius of the disk



And the feasibility problem Diffusion and environment

- ▶ Indirect density-dependence
- ▶ Source-sink dynamics
- ▶ Feasibility



Gravel et al. 2016. Nat. Communications

Conclusion



Stability and complexity Spatial dynamics

- ▶ Food webs are not random
- ▶ Spatial structure influence stability
- ▶ Distinct spatial rates leads to funky dynamics
- ▶ What's the value of this theory to the field ...

